

ATTENDANCE MANAGEMENT SYSTEM

Problems, Proposed Solutions and System Requirements

1. Purpose of the System

The Lecture Attendance Management System is designed to make lecture attendance easier, faster and more reliable. Students should be able to record attendance with very little effort, while the system automatically records and checks important attendance information. The lecturer should also be able to monitor attendance from the office without having to remain in the classroom.

2. Main Need

To make lecture attendance easier, faster, more accurate and more secure by automatically recording and checking student attendance.

The system is needed to reduce the time lecturers spend taking attendance manually, especially in large classes. It should make it easier to keep attendance records, check student attendance and calculate attendance percentages. It should also help reduce false attendance by checking the student's location when attendance is submitted.

3. Main Users

- Lecturers – create attendance sessions, generate QR codes, set the lecture location and monitor attendance.
- Students – scan the QR code and enter only their full name and registration number.
- Class Representatives – register students who genuinely do not have a phone, using a special authorised option.
- Academic Administrators – review attendance records when necessary.

4. Student Attendance Process

1. The lecturer creates an attendance session.
2. The system generates a unique QR code.
3. The lecturer displays the QR code in the lecture room.
4. A student scans the QR code.
5. The student enters only their full name and registration number.
6. The system automatically records the date and exact time.
7. The system automatically checks the student's location.
8. The system calculates the distance from the designated lecture location.
9. The system records the student as Present or Out of Range according to the location rule.
10. The attendance record becomes available to the lecturer on the dashboard.

5. Problem: Students Who Do Not Have a Phone

Some students may attend a lecture without a smartphone, may have a phone that cannot scan the QR code, or may have another genuine problem that prevents them from using the student attendance page. Without an alternative, these students may fail to record their attendance even though they are physically present.

6. Proposed Solution: Class Representative Registration

The system should provide a special registration option for an authorized class representative. The class representative can register two or more students who genuinely cannot use the QR code themselves. This option must not be available to ordinary students. Only an authorized class representative should be allowed to register attendance for multiple students.

7. Conditions for Class Representative Registration

1. Only an authorized class representative can use the multiple-student registration option.
2. The class representative must log in using their authorized account.
3. The normal attendance session must still be active.
4. The class representative enters the full name and registration number of each student.
5. The system records that the attendance was submitted by the class representative.
6. The system records the exact time of each registration.
7. The class representative can register two or more eligible students during the active session.
8. The lecturer can see which students registered themselves and which students were registered by the class representative.
9. The lecturer can review or remove an incorrect registration.
10. The class representative should not be allowed to register students after the 5-minute session has closed.

8. Attendance Methods

Method 1 – Normal Student Attendance

Scan QR code → Enter Full Name and Registration Number → Automatic time and location verification → Attendance recorded.

Method 2 – Class Representative Registration

Class Representative Login → Enter eligible student's Full Name and Registration Number → System records the class representative as the person who submitted the attendance → Attendance recorded.

9. Problem: Manual Attendance Takes Too Much Time

Calling names or using paper attendance lists can take a lot of lecture time, especially when there are many students. It can also result in missing names, incorrect records and difficulties when calculating attendance later.

10. Proposed Solution

Use a QR code to allow students to record attendance quickly. The system automatically stores the date, time, student details and attendance session information. This reduces manual work for the lecturer.

11. Problem: Proxy Attendance

A student may try to give the QR code to another person who is not in the lecture room, or students may attempt to register attendance for friends.

12. Proposed Solution

Use location verification together with the QR code. When a student submits attendance, the system checks the student's location and calculates the distance from the designated lecture location. Students outside the allowed area can be marked Out of Range or flagged for lecturer review.

13.Problem:

A student could scan or receive the QR code while they are at the hostel, home, or another place and try to register attendance without being in the lecture area. The lecturer may also be checking attendance from the office and therefore cannot physically confirm where the student was.

14. Proposed solution:

The system should use the student's GPS/location service when they submit attendance. It automatically compares the student's location with the designated lecture location.

For example, if the lecturer sets a 50-metre permitted radius:

Student within 50 meters → PRESENT

Student outside 50 meters → OUT OF RANGE

The system should automatically record:

Student's location

Lecture location

Distance between the two locations

Time of submission

Location verification result

Attendance status

Important condition

The student does not enter their location manually. The system requests location permission from the phone and captures the location automatically.

15. Problem: Attendance Code May Be Shared Later

If the same QR code remains active for a long time, a student could share it with another person who is not in the lecture.

16. Proposed Solution

Make each attendance QR code valid for exactly 5 minutes. After 5 minutes, the system automatically closes the attendance session and stops accepting new submissions.

17. Problem: Students Should Not Enter Technical Information

Students should not have to enter the date, time, GPS coordinates, distance or attendance status. Asking students to provide this information would make the system complicated and could allow false information.

18. Proposed Solution

Automate these processes. The student only enters Full Name and Registration Number. The system automatically handles the date, time, location, distance calculation, session validation and attendance status.

19. Data Recorded by the System

- Student Full Name.
- Student Registration Number.
- Course or Class.
- Attendance Session ID.
- Lecture Date.
- Exact Submission Time.
- Designated Lecture Location.
- Student Location, where location permission is available.
- Distance from the designated lecture location.
- Attendance Status.
- Attendance Method: Student or Class Representative.
- Class Representative identity, where applicable.
- Session Status.

20. Lecturer Dashboard

The lecturer dashboard should allow the lecturer to:

- Create a new attendance session.
- Set the course or class.
- Set the lecture location.

- Set the permitted location radius.
- Generate the QR code.
- Monitor attendance submissions.
- See students who are Present.
- See students who are Out of Range.
- See students registered by a class representative.
- Review attendance records.
- Calculate student attendance percentages.
- Close or review attendance sessions.

21. Security and Control

- Students should not have a dashboard or normal login.
- Only authorized class representatives should have the multiple-student registration option.
- The system must record who registered a student on behalf of another student.
- Class representative registration must be limited to the active 5-minute session.
- The lecturer should be able to review class representative registrations.
- Students should not manually enter their GPS location.
- The QR code should automatically become inactive after 5 minutes.

22. Complete System Flow

1. Lecturer opens the system and creates a lecture attendance session.
2. Lecturer sets the lecture location and permitted radius.
3. System generates a unique QR code.
4. The QR code is displayed in the classroom.
5. The 5-minute attendance period begins.
6. Students with phones scan the QR code and enter their details.
7. Students without phones can be registered by the authorized class representative.
8. The system automatically records time and performs location verification where applicable.
9. The system stores the attendance record and method used.
10. The lecturer views the attendance from the dashboard, including from the office.
11. After 5 minutes, the system automatically closes the attendance session.
12. The stored records are used to monitor attendance and calculate semester attendance percentages.

23. Key Principle

The main principle of the system is: the student should do as little as possible, while the system should do as much as possible automatically. Students normally only scan the QR code and enter their full name and registration number. For students who genuinely do not have a phone, an

authorized class representative provides the alternative registration method, with all such registrations clearly recorded for the lecturer to review.